

The α -Function

1. Introduction

This note introduces a concrete two-parameter family of functions defined by an integral transform

$$\alpha(x; a, b) := \int_0^\infty t^{a-1} (1+t)^{-b} e^{-xt} dt,$$

for suitable complex parameters a, b and real variable $x > 0$. The goal is modest:

1. Give a precise definition and convergence conditions.
2. Relate α to a standard special function (the confluent hypergeometric function U).
3. Exhibit explicit parameter choices that reproduce classical special functions such as the Gamma function, the Beta function, and the exponential integral.
4. Record a few basic identities useful for further analysis or computation.

Throughout, Γ denotes the Gamma function and B the Beta function.

2. Definition and basic properties

2.1 Definition

Definition 2.1 (The α -function). Let $x > 0$ and let $a, b \in \mathbb{C}$ satisfy $\operatorname{Re}(a) > 0$ and $\operatorname{Re}(b) > 0$. Define

$$\alpha(x; a, b) := \int_0^\infty t^{a-1} (1+t)^{-b} e^{-xt} dt.$$

Whenever the integral converges absolutely, this defines a complex-valued function of x that depends smoothly on the parameters a, b .

2.2 Convergence

Proposition 2.2 (Convergence for $x > 0$). If $x > 0$ and $\operatorname{Re}(a) > 0$, then for any fixed $b \in \mathbb{C}$ the integral defining $\alpha(x; a, b)$ converges absolutely.

Proof. Consider the two endpoints of integration.

- As $t \rightarrow 0^+$, we have $(1+t)^{-b} = 1 + O(t)$, so

$$t^{a-1} (1+t)^{-b} e^{-xt} \sim t^{a-1}.$$

The contribution near 0 is integrable if and only if $\operatorname{Re}(a) > 0$.

- As $t \rightarrow \infty$, we have $(1+t)^{-b} = t^{-b} (1+O(1/t))$, so

$$t^{a-1}(1+t)^{-b}e^{-xt} \sim t^{a-1-b}e^{-xt}.$$

The exponential factor e^{-xt} with $x > 0$ dominates any polynomial growth or decay, ensuring convergence at ∞ for all a, b .

Thus, for $x > 0$ and $\operatorname{Re}(a) > 0$, the integral converges absolutely. $\square$

Remark 2.3 (Analytic dependence). For fixed $x > 0$, the integrand is analytic in (a, b) on the region $\operatorname{Re}(a) > 0$, and the integral converges uniformly on compact subsets of this region. Hence $\alpha(x; a, b)$ is analytic in a and b on $\operatorname{Re}(a) > 0$ and admits analytic continuation beyond this domain via standard methods, if desired.

2.3 Basic identities

The integral representation makes some structural properties immediate.

Proposition 2.4 (Linearity). For fixed $x > 0$ and a , the map $b \mapsto \alpha(x; a, b)$ is analytic and linear with respect to the integrand $(1+t)^{-b}$; similarly for a .

This is just the linearity of the integral.

Proposition 2.5 (Differentiation with respect to x). For $x > 0$ and $\operatorname{Re}(a) > 0$,

$$\frac{\partial}{\partial x} \alpha(x; a, b) = -\alpha(x; a+1, b).$$

Proof. Differentiate under the integral sign (justified by absolute convergence and dominated convergence):

$$\frac{\partial}{\partial x} \alpha(x; a, b) = \frac{\partial}{\partial x} \int_0^\infty t^{a-1}(1+t)^{-b}e^{-xt} dt = \int_0^\infty t^{a-1}(1+t)^{-b} \frac{\partial}{\partial x} (e^{-xt}) dt = - \int_0^\infty t^a(1+t)^{-b}e^{-xt} dt = -\alpha(x; a+1, b).$$

$\square$

This identity gives a simple recursion in the parameter a when differentiating with respect to x .

3. Relation to the confluent hypergeometric function U

The confluent hypergeometric function of the second kind, $U(a, c, x)$, admits the integral representation

$$U(a, c, x) = \frac{1}{\Gamma(a)} \int_0^\infty e^{-xt} t^{a-1} (1+t)^{c-a-1} dt,$$

for $x > 0$ and $\operatorname{Re}(a) > 0$.

Comparing this with the definition of α , we obtain a simple parameter correspondence.

Proposition 3.1 (Identification with U). For $x > 0$ and $\operatorname{Re}(a) > 0$, we have

$$\alpha(x; a, b) = \Gamma(a) U(a, a + 1 - b, x).$$

Proof. Rewrite the exponent of $(1+t)$ in the definition of U :

$$(1+t)^{c-a-1} = (1+t)^{-b}$$

if and only if

$$c - a - 1 = -b \iff c = a + 1 - b.$$

Substituting $c = a + 1 - b$ into the integral representation of U gives

$$U(a, a + 1 - b, x) = \frac{1}{\Gamma(a)} \int_0^\infty e^{-xt} t^{a-1} (1+t)^{-b} dt,$$

so by the definition of α ,

$$\alpha(x; a, b) = \Gamma(a) U(a, a + 1 - b, x).$$

\qed

This identification immediately transfers many known properties of U to α .

Corollary 3.2 (Differential equation). For $x > 0$ and parameters with $\operatorname{Re}(a) > 0$, the function $x \mapsto \alpha(x; a, b)$ satisfies the confluent hypergeometric equation

$$x y''(x) + (a + 1 - b - x) y'(x) - a y(x) = 0.$$

Proof. The function $U(a, c, x)$ satisfies

$$x y''(x) + (c - x) y'(x) - a y(x) = 0.$$

With $c = a + 1 - b$ and $\alpha(x; a, b) = \Gamma(a) U(a, c, x)$, the same equation holds for α because multiplication by the constant factor $\Gamma(a)$ does not change the differential equation. \qed

4. Special cases

4.1 Gamma function

Setting $b=0$ removes the $(1+t)^{-b}$ factor.

Proposition 4.1 (Gamma as a special case). For $x > 0$ and $\operatorname{Re}(a) > 0$,

$$\alpha(x; a, 0) = \Gamma(a) x^{-a}.$$

Proof. With $b=0$ we have

$$\alpha(x; a, 0) = \int_0^\infty t^{a-1} e^{-xt} dt.$$

Make the substitution $u = xt$, so $t = u/x$ and $dt = du/x$:

$$\alpha(x; a, 0) = \int_0^\infty \left(\frac{u}{x}\right)^{a-1} e^{-u} \frac{du}{x} = x^{-a} \int_0^\infty u^{a-1} e^{-u} du = x^{-a} \Gamma(a).$$

\qed

Thus, apart from the elementary scaling factor x^{-a} , the Gamma function appears as the $b=0$ member of the α -family.

4.2 Beta function (limit $x \rightarrow 0^+$)

For $x=0$, the defining integral reduces to

$$\alpha(0; a, b) = \int_0^\infty t^{a-1} (1+t)^{-b} dt,$$

which converges under the additional condition $\operatorname{Re}(b) > \operatorname{Re}(a) > 0$. In this regime the integral can be expressed in terms of the Beta function.

Proposition 4.2 (Beta as a boundary value). If $\operatorname{Re}(b) > \operatorname{Re}(a) > 0$, then

$$\alpha(0; a, b) = B(a, b-a),$$

where

$$B(p, q) = \int_0^1 u^{p-1} (1-u)^{q-1} du.$$

Proof. Start with

$$\alpha(0; a, b) = \int_0^\infty t^{a-1} (1+t)^{-b} dt.$$

Make the substitution $t = \frac{u}{1-u}$, so that $u = \frac{t}{1+t}$, $dt = \frac{du}{(1-u)^2}$, and $1+t = \frac{1}{1-u}$. Then

$$\alpha(0; a, b) = \int_0^1 \left(\frac{u}{1-u}\right)^{a-1} \left(\frac{1}{1-u}\right)^{-b} \frac{du}{(1-u)^2} = \int_0^1 u^{a-1} (1-u)^{-(a-1)} (1-u)^b (1-u)^{-2} du = \int_0^1 u^{a-1} (1-u)^{b-a-1} du$$

The convergence condition $\operatorname{Re}(b) > \operatorname{Re}(a) > 0$ is exactly the usual condition for the Beta integral. \qed

Thus the Beta function appears as the $x \rightarrow 0^+$ boundary value of α in a suitable parameter range.

4.3 Exponential integral E_1

The exponential integral E_1 is defined for $x > 0$ by

$$E_1(x) = \int_x^\infty \frac{e^{-u}}{u} du.$$

Proposition 4.3 (Exponential integral as a special case). For $x > 0$,

$$\alpha(x; 1, 1) = e^x E_1(x).$$

Proof. For $a=b=1$ the definition gives

$$\alpha(x; 1, 1) = \int_0^\infty \frac{e^{-xt}}{1+t} dt.$$

Substitute $u = x(1+t)$, so that $t = u/x - 1$, $dt = du/x$, and $1+t = u/x$. Then when t runs from 0 to ∞ , u runs from x to ∞ , and

$$\alpha(x; 1, 1) = \int_{u=x}^\infty \frac{e^{-x(u/x-1)}}{u/x} \frac{du}{x} = \int_x^\infty e^x e^{-u} \frac{x}{u} \frac{du}{x} = e^x \int_x^\infty \frac{e^{-u}}{u} du = e^x E_1(x).$$

\qed

This exhibits another classical special function within the α -family.

5. Operator viewpoint and recursion

The definition of α may be viewed as a Laplace transform of a parameterized kernel. For fixed a, b , consider

$$K_{a,b}(t) = t^{a-1}(1+t)^{-b}, \quad t > 0.$$

Then

$$\alpha(x; a, b) = \mathcal{L}[K_{a,b}](x) := \int_0^\infty K_{a,b}(t) e^{-xt} dt.$$

This perspective emphasizes that:

- Many qualitative properties of α can be read off from $K_{a,b}$ (positivity for real parameters, decay, moments, etc.).

- Standard Laplace-transform techniques (inversion, convolution, Tauberian theorems) are available whenever $K_{a,b}$ satisfies the usual hypotheses.

The differentiation identity from Proposition 2.5 gives a simple recursion in a :

$$\frac{\partial}{\partial x} \alpha(x; a, b) = -\alpha(x; a + 1, b).$$

Repeated differentiation yields

$$\frac{\partial^n}{\partial x^n} \alpha(x; a, b) = (-1)^n \alpha(x; a + n, b), \quad n \in \mathbb{N}.$$

Together with the differential equation in Corollary 3.2, these relations provide a starting point for analytic and numerical work with α .

6. Summary

The function

$$\alpha(x; a, b) = \int_0^\infty t^{a-1} (1+t)^{-b} e^{-xt} dt$$

for $x > 0$ and $\operatorname{Re}(a) > 0$ is a concrete parametric family with the following features:

1. It is (up to a constant factor) the confluent hypergeometric function $U(a, a+1-b, x)$.
2. It recovers the Gamma function via $\alpha(x; a, 0) = \Gamma(a)x^{-a}$.
3. It recovers the Beta function as a boundary value: $\alpha(0; a, b) = B(a, b-a)$ under $\operatorname{Re}(b) > \operatorname{Re}(a) > 0$.
4. It contains the exponential integral E_1 as the special case $\alpha(x; 1, 1) = e^{-x}E_1(x)$.
5. It satisfies a simple recursion in a under differentiation with respect to x , and it solves the confluent hypergeometric differential equation.

These facts are sufficient to anchor the α -function in standard special-function theory and provide a clean base for any further applications or extensions.